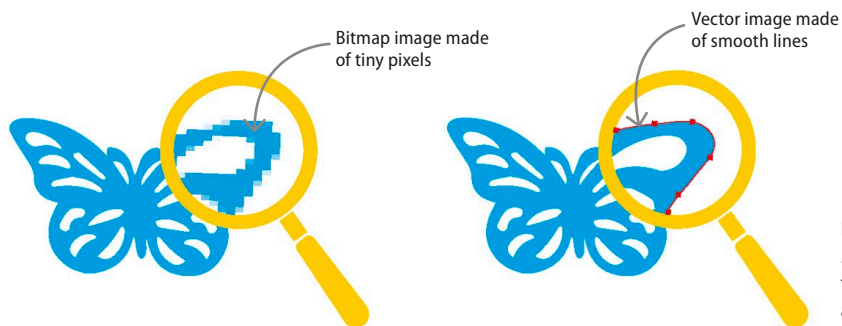


Encoding images

Images are encoded, or changed into a sequence of numbers, so that they can be understood by a computer. These images require different methods to store and display them.

Pixels and vectors

Images can be represented digitally by using two methods: bitmaps and vectors. Bitmaps are like mosaics made of lots of tiny squares called pixels, short for picture elements, arranged in a grid. Vector images are more like join-the-dots puzzles, as the image is represented by a collection of points, along with information on what sort of lines should join to make the shapes in the image.



SEE ALSO

◀ 40–41 How modern computers compute

◀ 80–81 Bits and digitization

◀ 82–83 Binary code

Encoding audio and video

92–93 ▶

IN DEPTH

Steganography

Steganography involves sending secret messages by hiding the fact that there is a message. This is like messages written in invisible ink. Information can be hidden in a digital image by using a program that changes only one bit of each binary number that represents the colours of the pixels. The same program can also extract a hidden message.

Formats

Digital cameras produce bitmapped images as .bmp, .jpeg, and .png files. Vector images, on the other hand, are created using illustration or animation programs and are stored as .svg files.

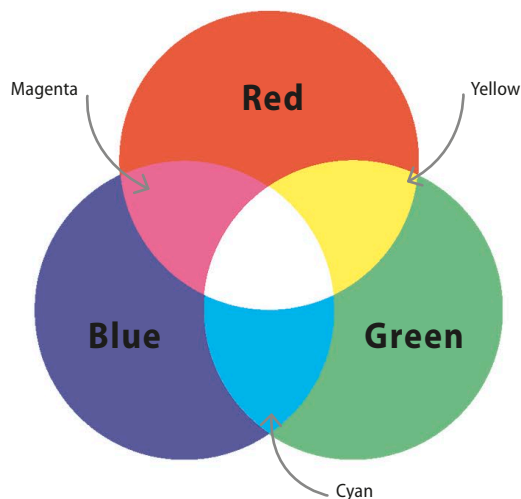
Colour and light

White light is made by mixing equal amounts of red, green, and blue light. Mixing these three primary colours in different proportions gives a wide range of other colours. An encoding called RGB (red, green, blue) is used to store colour values. An RGB value is written as three numbers between 0 and 255, each representing how much of the three primary colours it contains.

LINGO

Rasterization

Since computer screens are made up of a grid of pixels, vector images have to be turned into bitmaps before they can be displayed. This process is called rasterization, after the raster scan process used in older television screens to build up pictures a line at a time. The process of keeping the screen updated when users make changes involves a lot of work for the computer's processor and is sometimes done by a specialized chip.



△ Colour mixing

Black is 0, 0, 0 as it contains no light at all, and white is 255, 255, 255 – the maximum amount of all three colours. The rules of colour mixing for light are not the same as those for paints.